

◇ Real and Imaginary parts of a complex number:

Let $z = a + bi$. Then the **real and imaginary parts** of z are denoted as

$$\operatorname{Re}(z) = a \quad \text{and} \quad \operatorname{Im}(z) = b$$

◇ **Euler's Formula:** $e^{i\theta} = \cos \theta + i \sin \theta$

◇ These ideas lead to the following very important identities for $\cos \theta$ and $\sin \theta$.

$$\cos \theta = \operatorname{Re}(e^{i\theta}) = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \operatorname{Im}(e^{i\theta}) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

◇ Euler's formula gives another way to view the polar form of a complex number.

Recall that $z = a + bi$ can be written in the following **polar form**

$$z = r(\cos \theta + i \sin \theta)$$

where $a = r \cos \theta$, $b = r \sin \theta$, and $r = |z| = \sqrt{a^2 + b^2}$

In terms of Euler's formula, any complex number can therefore be written in the form

$$z = |z|e^{i\theta}$$

◇ Useful trig identities

$$\cos(-\theta) = \cos(\theta) \quad \text{and} \quad \sin(-\theta) = -\sin(\theta)$$

Example 1: Write each of the following complex numbers in standard form ($a + bi$)

(a) $e^{-i\pi/4}$

$$\begin{aligned} \text{Solution : } e^{-i\pi/4} &= e^{i(-\pi/4)} \\ &= \cos(-\pi/4) + i \sin(-\pi/4) \end{aligned}$$

(b) $e^{2e^{3+i\pi/6}}$

$$\begin{aligned} \text{Solution : } 2e^{3+i\pi/6} &= 2e^3 e^{i\pi/6} \\ &= 2e^3 [\cos(\pi/6) + i \sin(\pi/6)] \\ &= 2e^3 \cos(\pi/6) + i 2e^3 \sin(\pi/6) \end{aligned}$$

(c) e^{e^i}

$$\begin{aligned} \text{Solution : } e^{e^i} &= e^{\cos(1) + i \sin(1)} \\ &= e^{\cos(1)} e^{i \sin(1)} \\ &= e^{\cos(1)} [\cos(\sin(1)) + i \sin(\sin(1))] \\ &= e^{\cos(1)} \cos(\sin(1)) + i e^{\cos(1)} \sin(\sin(1)) \end{aligned}$$

$$(d) \frac{e^{3i} - e^{-3i}}{2i}$$

$$\begin{aligned} \text{Solution : } \frac{e^{3i} - e^{-3i}}{2i} &= \frac{(\cos(3) + i \sin(3)) - (\cos(-3) + i \sin(-3))}{2i} \\ &= \frac{[\cos(3) + i \sin(3)] - [\cos(3) - i \sin(3)]}{2i} \\ &= \frac{2i \sin(3)}{2i} \\ &= \sin(3) = \sin(3) + 0i \end{aligned}$$

Example 2: Write each of the following complex numbers in polar form ($re^{i\theta}$)

$$(a) \frac{1-i}{3}$$

Solution : Note this point would be in the 4th quadrant

$$\begin{aligned} r &= \left| \frac{1-i}{3} \right| = \left| \frac{1}{3} - \frac{i}{3} \right| = \sqrt{\frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3} \\ \theta &= \tan^{-1} \left(\frac{\frac{1}{3}}{-\frac{1}{3}} \right) = \tan^{-1}(-1) = \frac{3\pi}{4} \text{ or } -\frac{\pi}{4} \end{aligned}$$

Since θ must be in the 4th quadrant $\implies \theta = -\frac{\pi}{4}$

$$\text{Therefore } \frac{1-i}{3} = \frac{\sqrt{2}}{3} e^{-i\pi/4}$$

$$(b) (1+i)^6$$

Solution : Lets convert $1+i$ first. (It is in the 1st quadrant)

$$\begin{aligned} r &= |1+i| = \sqrt{1+1} = \sqrt{2} \\ \theta &= \tan^{-1} \left(\frac{1}{1} \right) = \tan^{-1}(1) = \frac{\pi}{4} \text{ or } -\frac{\pi}{4} \end{aligned}$$

Since θ must be in the 1st quadrant $\implies \theta = \frac{\pi}{4}$

$$\text{Therefore } (1+i)^6 = \left(\sqrt{2} e^{i\pi/4} \right)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2}$$

$$(c) (1+i)^{1/2}$$

Solution : From (b), we have $1 + i = \sqrt{2}e^{i\pi/4}$

$$\text{Therefore } (1 + i)^{1/2} = \left(2^{1/2}e^{i\pi/4}\right)^{1/2} = 2^{1/4} \left(e^{i\pi/4}\right)^{1/2} = 2^{1/4}e^{i(\pi/8+2\pi k/2)} \quad k = 0, 1$$

$$k = 0 : \quad 2^{1/4}e^{i(\pi/8)}$$

$$k = 1 : \quad 2^{1/4}e^{i(9\pi/8)}$$

(d) $\frac{2i}{3e^{4+i}}$

$$\text{Solution : } \frac{2i}{3e^{4+i}} = \frac{2e^{i\pi/2}}{3e^4e^i} = \frac{2}{3e^4}e^{i\pi/2-i} = \frac{2}{3e^4}e^{i(\pi/2-1)}$$

Exercises

Using Euler's polar form only (no sines or cosines), compute the following complex powers and leave your final answer in Euler's polar form.

1. $(2 + 2i)^6$
2. $(2 + 2i)^{1/3}$
3. $4(1 - \sqrt{3}i)^7$
4. $4(1 - \sqrt{3}i)^{1/4}$
5. $\left(\frac{-2 + 2i}{3i}\right)^{1/2}$ (Hint: Write the quotient in Euler's form first)

Find all complex solutions of the following equations and leave your answers in Euler's polar form

6. $z^{1/6} - 2 - 2i = 0$
7. $z^3 + 5i = 2 + 7i$
8. $3iz^2 + 2 = 2i$